

ASSIGNMENT 9.1

Name: P.Niharika

Ht.No: 2303A51694

Batch: 24

Problem 1:

Consider the following Python function:

```
def find_max(numbers):  
    return max(numbers)
```

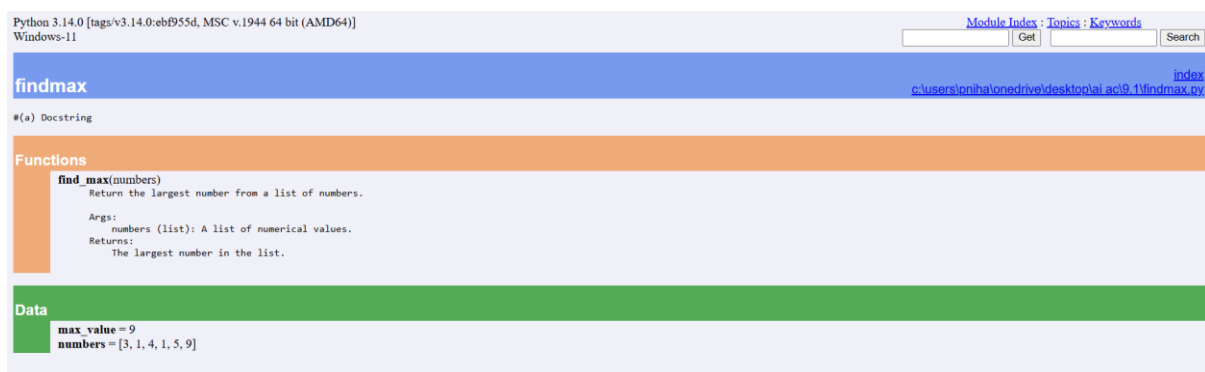
Task:

- Write documentation for the function in all three formats:
 - (a) Docstring
 - (b) Inline comments
 - (c) Google-style documentation
- Critically compare the three approaches. Discuss the advantages, disadvantages, and suitable use cases of each style.
- Recommend which documentation style is most effective for a mathematical utilities library and justify your

answer.

```
1 # (a) Docstring
2 def find_max(numbers):
3     """
4     Return the largest number from a list of numbers.
5     """
6     return max(numbers)
7 # (b) Inline Comments
8 def find_max(numbers):
9     # Use the built-in max function to find the highest value in the sequence
10    return max(numbers) # Returns the maximum value found
11 # (c) Google-Style Documentation
12 def find_max(numbers):
13    """
14    Return the largest number from a list of numbers.
15
16    Args:
17    | numbers (list): A list of numerical values.
18    Returns:
19    | The largest number in the list.
20    """
21    return max(numbers)
22 numbers = [3, 1, 4, 1, 5, 9]
23 max_value = find_max(numbers)
24 print(max_value) # Output: 9
25
```

9
PS C:\Users\pniha\OneDrive\Desktop\AI AC\9.1> py -m pydoc -p 8080
Server ready at http://localhost:8080/
Server commands: [b]rowser, [q]uit
server> b
server>



Problem 2: Consider the following Python function:

```
def login(user, password, credentials):
```

```
    return credentials.get(user) == password
```

Task:

1. Write documentation in all three formats.
2. Critically compare the approaches.
3. Recommend which style would be most helpful for new developers onboarding a project, and justify your choice.

```
Untitled-1 • findmax.py • credentials.py X
credentials.py > ...
1 #1)Doctsring
2 def login(user,password,credentials):
3     """This function checks if the provided username and password match the credentials stored in a dictionary.
4     If the credentials are valid, it returns a success message. Otherwise, it raises a ValueError with an error message.
5
6     """
7     if user in credentials and credentials[user] == password:
8         return "Login successful"
9     else:
10        raise ValueError("Invalid username or password")
11 #2)inline suggestions
12 def login(user,password,credentials): #to find the user in credentials and check if the password matches
13     if user in credentials and credentials[user] == password:#if the user is found and the password matches, return a success message
14         return "Login successful"
15     else: #if the user is not found or the password does not match, raise a ValueError with an error message
16         raise ValueError("Invalid username or password")
17 #3)google style docstring
18 def login(user,password,credentials):
19     """_summary_
20
21     Args:
22         user (_type_): _description_
23         password (_type_): _description_
24         credentials (_type_): _description_
25     returns:
26         _type_: _description_
27     raises:
28         ValueError: _description_
29     """
30     if user in credentials and credentials[user] == password:
31         return "Login successful"
32     else:
33         raise ValueError("Invalid username or password")
34 user="admin"
35 password="admin123"
36 credentials={"admin":"admin123","user1":"password1"}

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\pnaha\OneDrive\Desktop\AI AC\9.1> py -m pydoc -p 8080
Server ready at http://localhost:8080/
Server commands: [b]rowser, [q]uit
server> b
```

Python 3.14.0 [tags/v3.14.0:eb955d, MSC v.1944 64 bit (AMD64)]
Windows-11

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credentials

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#1)Doctsring

Functions

```
login(user, password, credentials)
    _summary_

    Args:
        user (_type_): _description_
        password (_type_): _description_
        credentials (_type_): _description_
    returns:
        _type_: _description_
    raises:
        ValueError: _description_
```

Data

```
credentials = {'admin': 'admin123', 'user1': 'password1'}
password = 'admin123'
user = 'admin'
```

Problem 3: Calculator (Automatic Documentation Generation)

Task: Design a Python module named calculator.py and demonstrate automatic documentation generation.

Instructions:

1. Create a Python module calculator.py that includes the following functions, each written with appropriate docstrings:
 - o add(a, b) – returns the sum of two numbers
 - o subtract(a, b) – returns the difference of two numbers
 - o multiply(a, b) – returns the product of two numbers

- o `divide(a, b)` – returns the quotient of two numbers
2. Display the module documentation in the terminal using Python's documentation tools.
 3. Generate and export the module documentation in HTML format using the `pydoc` utility, and open the generated HTML file in a web browser to verify the output.

```

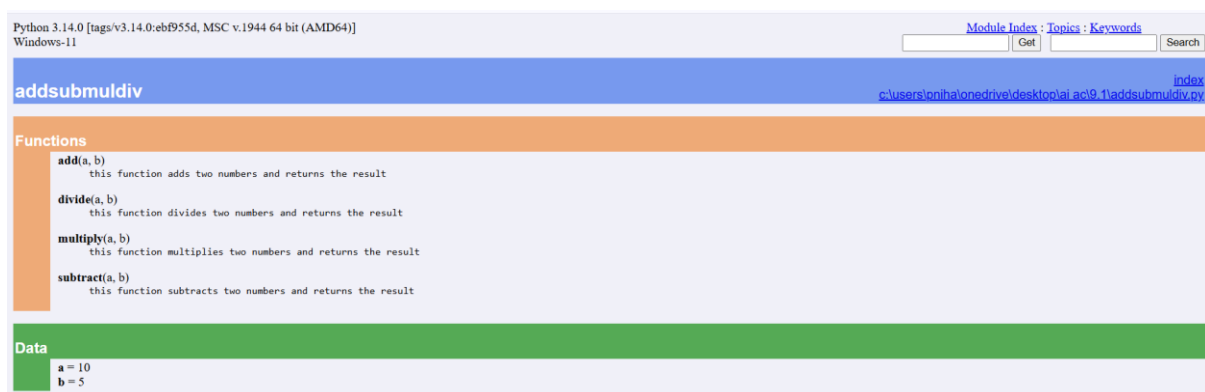
addsubmuldiv.py > ...
1  def add(a,b):
2      """this function adds two numbers and returns the result
3          """
4      return a+b
5  def subtract(a,b):
6      """this function subtracts two numbers and returns the result
7          """
8      return a-b
9  def multiply(a,b):
10     """this function multiplies two numbers and returns the result
11         """
12     return a*b
13 def divide(a,b):
14     """this function divides two numbers and returns the result
15         """
16     if b==0:
17         return "Error: Division by zero is not allowed."
18     return a/b
19 a=10
20 b=5
21 print("Addition:", add(a,b))
22 print("Subtraction:", subtract(a,b))
23 print("Multiplication:", multiply(a,b))
24 print("Division:", divide(a,b))

```

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Addition: 15
Subtraction: 5
Multiplication: 50
Division: 2.0
PS C:\Users\pnaha\OneDrive\Desktop\AI AC\9.1> py -m pydoc -p 8080
Server ready at http://localhost:8080/
Server commands: [b]rowser, [q]uit
server> b

```



Problem 4: Conversion Utilities Module

Task:

1. Write a module named `conversion.py` with functions:
 - o `decimal_to_binary(n)`

o `binary_to_decimal(b)`

o `decimal_to_hexadecimal(n)`

2. Use Copilot for auto-generating docstrings.

3. Generate documentation in the terminal.

4. Export the documentation in HTML format and open it in a browser.

```
Untitled-1 • findmax.py • credentials.py • addsubmuldiv.py • decimal to binary.py X
decimal to binary.py > ...
1 def decimal_to_binary(n):
2     """this function converts a decimal number to binary and returns it as a string
3     """
4     if n == 0:
5         return "0"
6     binary = ""
7     while n > 0:
8         binary = str(n % 2) + binary
9         n = n // 2
10    return binary
11
12 n=6
13 print(decimal_to_binary(n))
14 def binary_to_decimal(b):
15     """this function converts a binary number (given as a string) to decimal and returns it as an integer
16     """
17     decimal = 0
18     for i in range(len(b)):
19         decimal += int(b[-(i+1)]) * (2 ** i)
20    return decimal
21 b="110"
22 print(binary_to_decimal(b))
23 def decimal_to_hexadecimal(n):
24     """this function converts a decimal number to hexadecimal and returns it as a string
25     """
26     if n == 0:
27         return "0"
28     hexadecimal = ""
29     while n > 0:
30         remainder = n % 16
31         if remainder < 10:
32             hexadecimal = str(remainder) + hexadecimal
33         else:
34             hexadecimal = chr(remainder - 10 + ord('A')) + hexadecimal
35         n = n // 16
36    return hexadecimal
37 n=225
38 print(decimal_to_hexadecimal(n))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\priha\OneDrive\Desktop\AI AC\9.1> & C:\Users\priha\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/priha/OneDrive/Desktop/AI AC/9.1/decimal to binary.py"
6
E1
PS C:\Users\priha\OneDrive\Desktop\AI AC\9.1> py -m pydoc -p 8888
Server ready at http://localhost:8888/
Server commands: [b]rowser, [q]uit
server> h
```

calclater

Functions

`binary_to_decimal(b)`

this function converts a binary number (given as a string) to decimal and returns it as an integer

`decimal_to_binary(n)`

this function converts a decimal number to binary and returns it as a string

`dicimal_to_hexadecimal(n)`

this function converts a decimal number to hexadecimal and returns it as a string

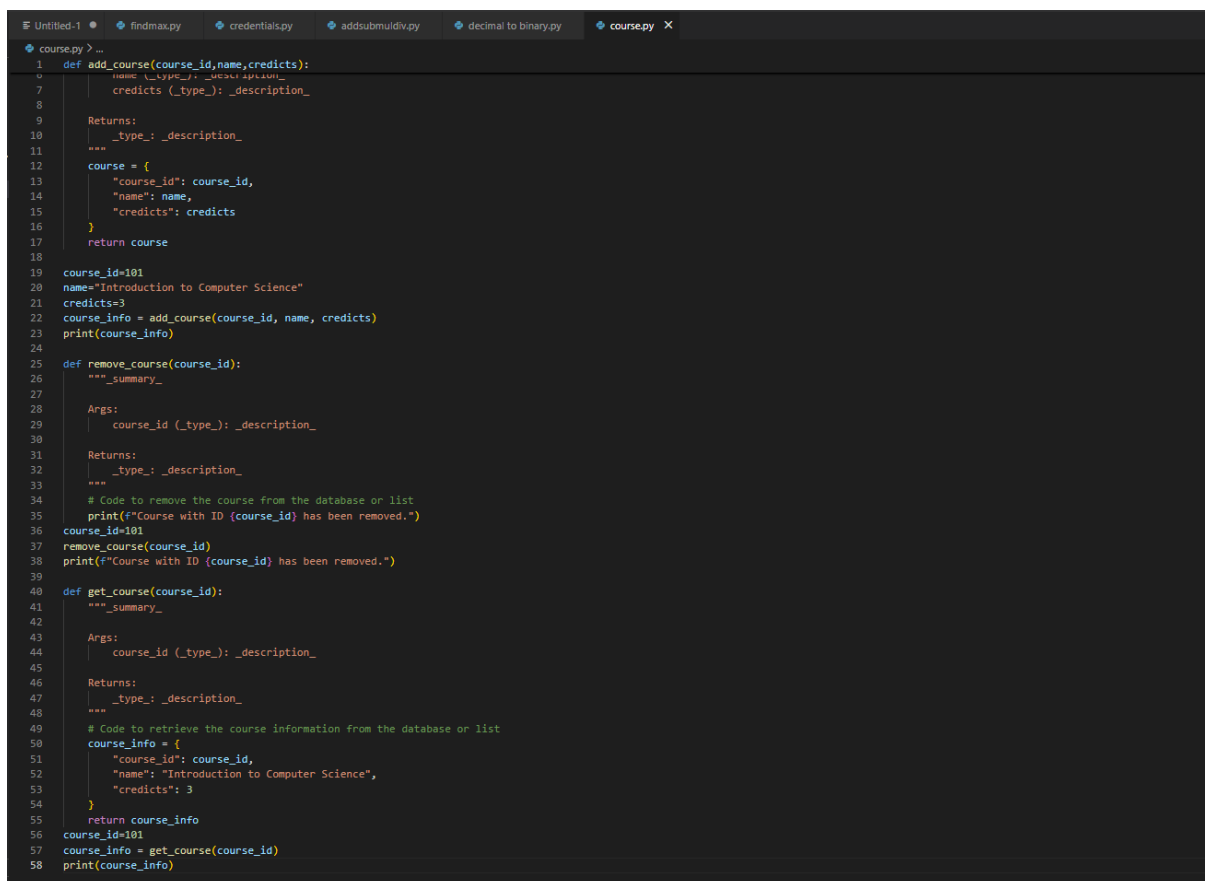
Data

```
b = '110'
n = 225
```

Problem 5 – Course Management Module

Task:

1. Create a module `course.py` with functions:
 - o `add_course(course_id, name, credits)`
 - o `remove_course(course_id)`
 - o `get_course(course_id)`
2. Add docstrings with Copilot.
3. Generate documentation in the terminal.
4. Export the documentation in HTML format and open it in a browser.



```
1 def add_course(course_id,name,credits):
2     """
3     name (_type_): _description_
4     credits (_type_): _description_
5
6     Returns:
7         _type_: _description_
8     """
9     course = {
10         "course_id": course_id,
11         "name": name,
12         "credits": credits
13     }
14     return course
15
16 course_id=101
17 name="Introduction to Computer Science"
18 credits=3
19 course_info = add_course(course_id, name, credits)
20 print(course_info)
21
22 def remove_course(course_id):
23     """
24     Args:
25         course_id (_type_): _description_
26
27     Returns:
28         _type_: _description_
29     """
30     # Code to remove the course from the database or list
31     print(f"Course with ID {course_id} has been removed.")
32     course_id=101
33     remove_course(course_id)
34     print(f"Course with ID {course_id} has been removed.")
35
36 def get_course(course_id):
37     """
38     Args:
39         course_id (_type_): _description_
40
41     Returns:
42         _type_: _description_
43     """
44     # Code to retrieve the course information from the database or list
45     course_info = {
46         "course_id": course_id,
47         "name": "Introduction to Computer Science",
48         "credits": 3
49     }
50     return course_info
51
52 course_id=101
53 course_info = get_course(course_id)
54 print(course_info)
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\pniha\OneDrive\Desktop\AI AC\9.1> & C:\Users\pniha\AppData\Local\Python\pythoncore-3.14-64\python
{'course_id': 101, 'name': 'Introduction to Computer Science', 'credits': 3}
{'course_id': 101, 'name': 'Introduction to Computer Science', 'credits': 3}
Course with ID 101 has been removed.
Course with ID 101 has been removed.
{'course_id': 101, 'name': 'Introduction to Computer Science', 'credits': 3}
PS C:\Users\pniha\OneDrive\Desktop\AI AC\9.1> py -m pydoc -p 8080
PS C:\Users\pniha\OneDrive\Desktop\AI AC\9.1> py -m pydoc -p 8080
Server ready at http://localhost:8080/
Server commands: [b]rowser, [q]uit
server> b
```

Python 3.14.0 [tags/v3.14.0:ebf955d, MSC v.1944 64 bit (AMD64)]
Windows-11

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[c:\users\pniha\onedrive\desktop\ai ac\9.1\course.py](#)

Functions

```
add_course(course_id, name, credits)
    _summary_

    Args:
        course_id (_type_): _description_
        name (_type_): _description_
        credits (_type_): _description_

    Returns:
        _type_: _description_

get_course(course_id)
    _summary_

    Args:
        course_id (_type_): _description_

    Returns:
        _type_: _description_

remove_course(course_id)
    _summary_

    Args:
        course_id (_type_): _description_

    Returns:
        _type_: _description_
```

Data

```
course_id = 101
course_info = {'course_id': 101, 'credits': 3, 'name': 'Introduction to Computer Science'}
credits = 3
name = 'Introduction to Computer Science'
```